Enhancing *Plasmodium* Research through plasmoRUtils: A one-stop R Package for Apicomplexan Biology

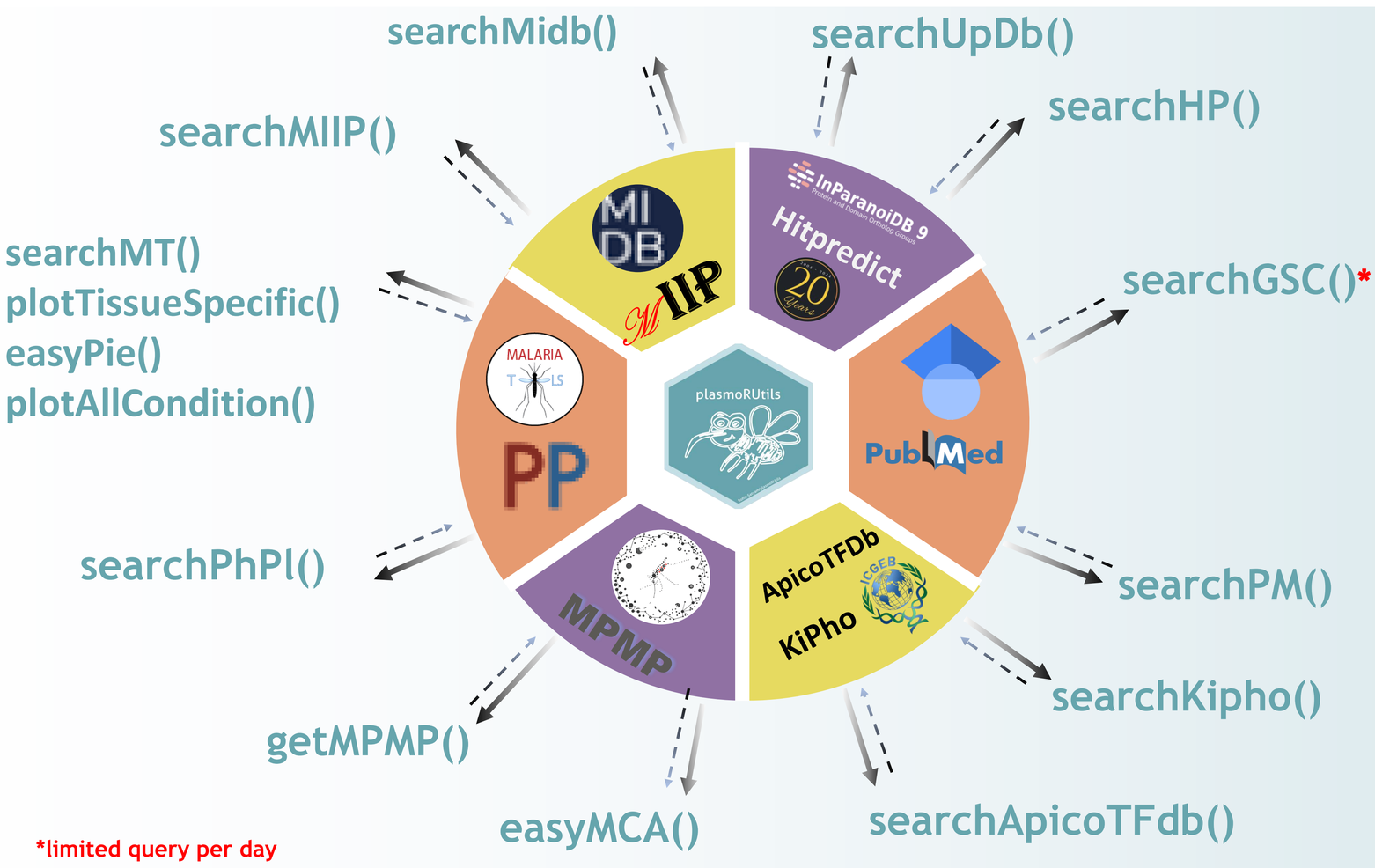
Rohit Satyam[1], Arnab Pain[1]

[1]Pathogen Genomics Group, Bioscience Program, Biological and Environmental Science and Engineering (BESE) Division, King Abdullah University of Science and Technology, Thuwal, Kingdom of Saudi Arabia

#1 Motivation

1. Handle outdated gene IDs , poor quality plots, discontinued advance analysis & visualization services.
2. To automate highly redundant manual but advanced task & develop CLI to access database without leaving RStudio.
3. Multiple independent *Plasmodium* annotation resources lack APIs and don't communicate with each other.
4. Datasets were too large to download, reanalyse, revise, reproduce, reuse & required bioinformatics expertise.
5. Develop support for batch queries, crawl multiple hyperlinks to access, sanitize & export most updated data whenever required.

#2 Highlights

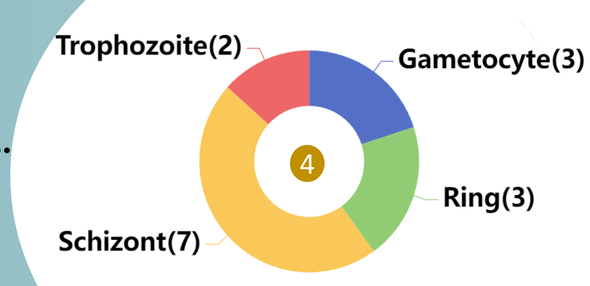


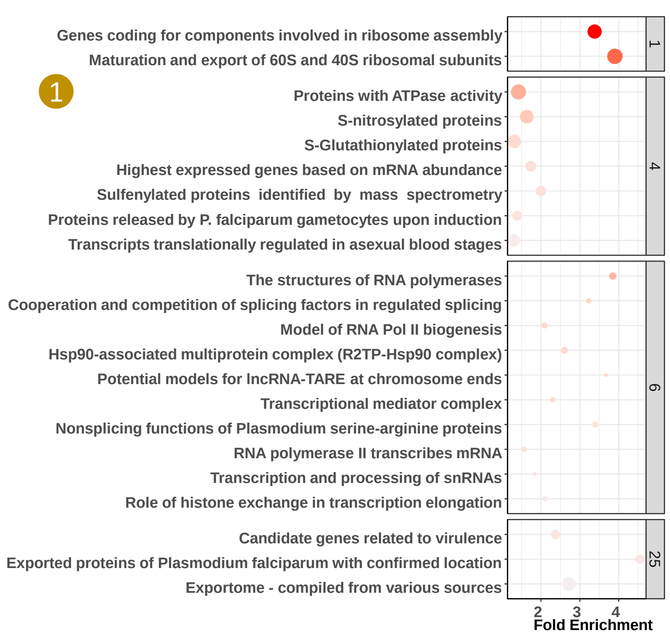
Given a list of Gene IDs user can

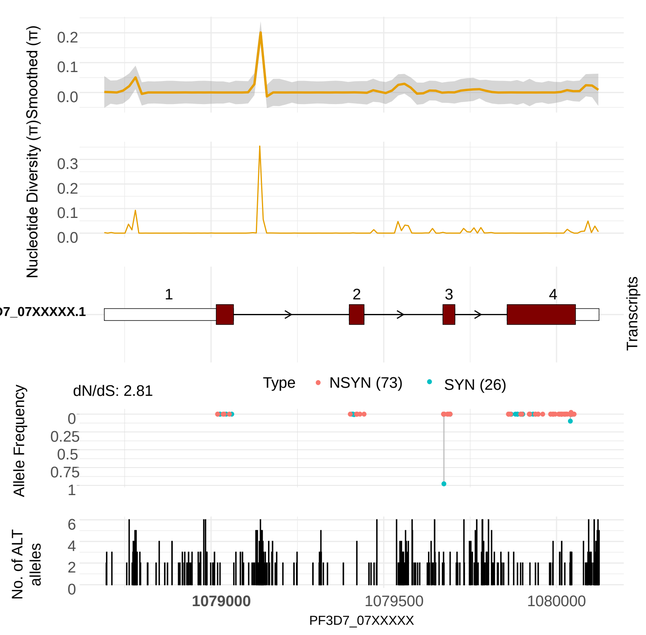
- Query multiple database to get information such as PPIs, articles published, Phenotype, Orthologs, Stage Specific expression values, etc. quickly without leaving RStudio.
- Create complex & publication ready visualizations effortlessly.
- Carry out routine tasks such as GO Enrichment analysis using custom GAF files, pathway enrichment analysis using updated MPMP geneset, Stage and Timepoint QC of expression data etc.
- Have access to reanalysed and annotated *Plasmodium* datasets such as Bulk & scRNASeq, population diversity statistics from Pf7K, functions assigned to hypothetical proteins, GRNs and high confidence PPI network.
- Detailed vignettes on how to use other R packages with plasmoRUtils to assist data analysis and exploration.

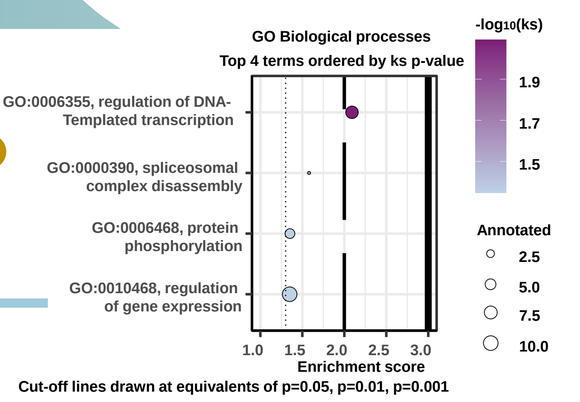
#3 Exemplary Visualization

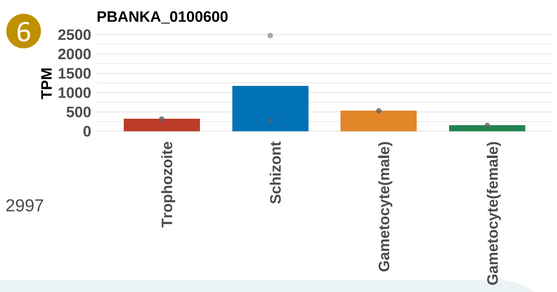
1. > easyPFR(gset=mpmp_gset): Runs pathfindeR using MPMP dataset.
   > plotPFR(): Quick visualization of Pathway enrichment results.
2. > plotPf7k(): Plot Quick genic diversity and summary statistics for any gene using Pf7k dataset.
3. > plotDomain(): Quickly plot various domains present for your list of proteins. Domain information is retrieved from Uniprot.
4. >easyPie(): Quickly plot stages in which genes are expressed. Require output of searchMT() function.
5. > easyGOPlot(): Quickly plot results of easytopGO() function.
6. > plotTissueSpecific(): Quickly fetches data from malaria.tools and reproduce the high quality plots similar to the server. Can be scaled for multiple PfIDs.











Acknowledgements: My sincere thanks to the #EuroBioC2024 & Bioconductor Team for developmental support and PlasmoDB for providing necessary data and support. This work is funded by Faculty Baseline Fund of Prof. Arnab Pain (BAS/1/1020-01-01).

References: The plasmoRUtils depends on the following R packages. For detailed references, please scan the QR Code

Bioconductor — OPEN SOURCE SOFTWARE FOR BIOINFORMATICS

PlasmoDB — Plasmodium Informatics Resources

EuroBioC2024

UniprotR · pathfindR · rvest · tidyverse · pRoloc · biomaRt · ggplot2 · drawProteins

King Abdullah University of Science and Technology